# My Google Interview Cheat Sheet

*One page. Glance at this the morning of an interview — not for studying, just for recall triggers.*

| # | Category | Pri | Core Pattern | Most-Likely Google Problems |
|---|---|---|---|---|
| 1 | **Two Pointers** | **HIGH** | Converge from ends, or slow-builds + fast-scans | Two Sum, 3Sum, Container With Most Water |
| 2 | **Sliding Window** | **HIGH** | Expand right; shrink left when invalid | Longest Substring w/o Repeat, Min Window Substring |
| 3 | **Prefix Sum / HashMap** | **HIGH** | Precompute once → O(1) lookups after | Product Except Self, Longest Consecutive Sequence |
| 4 | **Binary Search** | **HIGH** | Halve space — classic / rotated / on-the-answer | Median of Two Sorted Arrays, Search Rotated Array |
| 5 | **Intervals** | **HIGH** | Sort by start (merge) / end (greedy) / heap (concurrent) | Insert Interval, Meeting Rooms II, Employee Free Time |
| 6 | **Fast & Slow Pointers** | **MED** | slow +1, fast +2 | Linked List Cycle, Find the Duplicate Number |
| 7 | **Tree DFS/BFS** | **HIGH** | Bottom-up / top-down / level-order / BST-exploit / serialize | Serialize & Deserialize BT, Validate BST, LCA of BST |
| 8 | **Graphs** | **HIGH** | BFS=shortest path · DFS=cycle/components · Kahn=topo sort · Union-Find=incremental | Number of Islands, Course Schedule, Alien Dictionary |
| 9 | **Tries** | **MED** | children map + is_end flag | Implement Trie, Word Search II, Add and Search Words |
| 10 | **1D DP** | **HIGH** | Fibonacci / linear-decision / Kadane / LIS-scan | Word Break, House Robber, Maximum Subarray |
| 11 | **2D DP** | LOW | Two-sequence align, or grid-path sum from neighbors | Longest Common Subsequence, Unique Paths |
| 12 | **Backtracking** | **MED** | Include-or-skip / used-tracking / grid DFS+undo | Word Search, Combination Sum |
| 13 | **Heap / Top-K** | **MED** | Size-K heap · two balanced heaps · k-way merge · greedy sim | Top K Frequent Elements, Find Median Data Stream, Task Scheduler |
| 14 | **Stack-Based** | **MED** | Match-and-resolve, or monotonic increasing/decreasing | Valid Parentheses, Sliding Window Maximum, Asteroid Collision |
| 15 | **Matrix Traversal** | LOW | Boundary-shrink (spiral) / transpose+reverse (rotate) | Spiral Matrix, Rotate Image, Set Matrix Zeroes |
| 16 | **Bit Manipulation** | LOW | XOR cancels pairs · n&(n-1) clears lowest bit | Missing Number, Sum of Two Integers, Counting Bits |
| 17 | **Design a Data Structure** | **HIGH** | HashMap + (linked list / freq buckets / swap-pop array / sorted list) | LRU Cache, Time Based KV Store, Insert Delete GetRandom |

***Pri legend:*** *HIGH (dark shading) = must-know, will almost certainly appear · MED (light shading) = important, very common · LOW (white) = good-to-know, lower frequency*

### Before you touch code, every time:

1. Restate the problem in your own words. Ask about edge cases (empty input, duplicates, negatives).
2. Say the brute force out loud first, with its complexity — even when you already see the better approach.
3. State your chosen approach + its time/space complexity before writing it.
4. After coding: trace through one example by hand before saying “done.”

*If stuck more than 10 min with zero progress — that's signal, not failure. Narrate where you're stuck; interviewers grade the thinking, not just the destination.*